

Sharath H N

2nd Year MS (Research), CSE, Indian Institute of Technology Bombay
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Research Interests

Mechanistic interpretability, studying circuits and representations in language models. Focus on encoding of non-literal, figurative, and culturally grounded semantics.

Education

Indian Institute of Technology Bombay

Master of Science (by Research) in Computer Science and Engineering

MH, India

2024 – 2027 (expected)

- **GPA:** 9.31 / 10
- **Focus:** Understanding Figurative Language through Representation Steering
- **Advisors:** Prof. Pushpak Bhattacharya; Prof. Swaprava Nath; Prof. Aditya Joshi (UNSW, Australia)

JSS Science and Technology University, Mysuru

Bachelor of Engineering in Computer Science and Engineering

Karnataka, India

2018 – 2022

- **GPA:** 9.51 / 10
- **Thesis:** Impact of Different Embeddings on Text Classification
- **Advisors:** Prof. Anilkumar K.M

Publications

1. **Sharath Naganna**, Tanvir Ahmed Sijan, Uddipta Kalita. *“Are Arithmetic Heuristic Neurons Form-Invariant? A Mechanistic Analysis of Symbols, Text, and Code in LLMs.”* Under review. arXiv:2607.16693
2. **Sharath Naganna**, Ritesh Sur Chowdhury, Mrinal Devnath, Abhinav Joshi, Aditya Joshi, Ashutosh Modi. *“The Underlying Mechanistic Connection between Zero-Shot and In-Context Learning.”* Under review.
3. **Sharath Naganna**, Ritesh Sur Chowdhury, Dipankar Srirag, Aditya Joshi. *“Scaled Autoregressive Projection Steering (SAPS): Token-level Steering to Induce Sarcasm and Humblebrag in Text.”* Under review.
4. Akshata A. Bhat, **Sharath Naganna**, Saiful Haq, Prashant Khatri, Krishna Chaitanya Reddy Tamataam, Neha Arun, Niyati Chhaya, Pushpak Bhattacharya. *“SAVIOR: Sample-efficient Adaptation of Vision-Language Models for OCR Representation.”* Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops, 2026, pp. 719–728.
5. **Sharath Naganna**, Saprativa Bhattacharjee, Biplab Banerjee, Pushpak Bhattacharyya. *“My life is miserable, have to sign 500 autographs everyday”: Exposing Humblebragging, the Brags in Disguise.* accepted at Findings of the Association for Computational Linguistics: ACL 2025, Vienna, Austria

Research Experience

MS (Research) Thesis: Representation-Level Control of Figurative Language in Large Language Models

Indian Institute of Technology, Bombay (Prof. Aditya Joshi (UNSW, Australia))

2025–present

- Studying representation-level control of figurative language (sarcasm and humblebragging) in large language models using activation steering and mechanistic interpretability techniques.
- Proposed *Autoregressive Projection Steering (APS)*, a dynamic residual-stream intervention that re-computes steering coefficients during generation to improve stability and linguistic fluency.
- Performed mechanistic analysis of steering dynamics through layer-wise activation studies and token-level distribution divergence, demonstrating improved stability of APS compared to prior steering methods.

Mechanistic Connection between Zero-Shot and In-Context Learning

Indian Institute of Technology, Kanpur (Prof. Ashutosh Modi)

2025–present

- Compared the sparse computational circuits underlying **zero-shot** and **in-context learning (ICL)** using **edge attribution patching with integrated gradients (EAP-IG)** across six tasks and three model families (**Llama-3.1-8B**, **Mistral-7B**, **Qwen2.5-7B**).

- Measured circuit compressibility, cross-regime transfer, and structural overlap, finding **asymmetric transfer**: ICL circuits (median **5.95× larger**) preserve zero-shot behavior, while zero-shot circuits fail to recover ICL behavior.
- Established that a median **86% of zero-shot circuit edges** are contained within the corresponding ICL circuit (**3.59× above chance**), evidencing that ICL **extends rather than replaces** the zero-shot task-solving computation.

Sample-efficient Adaptation of Vision-Language Models for OCR Representation

Hyperbots, Bengaluru (Supervisor: Dr. Niyati Chhaya)

May 2025–July 2025

- Proposed **SAVIOR**, a sample-efficient data curation strategy to adapt vision–language models for robust financial OCR under real-world failure modes.
- Built **SAVIOR-Train** (2,234 docs) and **SAVIOR-Bench** (509 expert-annotated docs) targeting underrepresented OCR challenges such as vertical text, fine print, and complex layouts.
- Introduced **PaIRS**, a structure-aware metric measuring pairwise spatial relationships between tokens to evaluate layout fidelity beyond word accuracy.
- Fine-tuned Qwen2.5-VL with LoRA, achieving **0.93 word recall** and **0.89 F1** on downstream document QA.

Projects

Form-Invariance of Arithmetic Circuits in LLMs

2026

Mechanistic Interpretability of Arithmetic Reasoning in LLMs

[code]

- Investigated whether arithmetic “**heuristic neurons**” are **form-invariant** across symbolic arithmetic, natural-language word problems, and Python code in three **Llama-3** models.
- Localized these neurons using a two-stage pipeline combining **attribution patching** and **activation patching**, applied separately to each input format.
- Showed a **compact shared neuron set** across all three formats, with targeted interventions confirming the circuit is **both necessary and sufficient** for late-layer arithmetic computation.

Multi-model Indian Humor Summarizer

Fall 2025

Course: Speech, Natural Language Processing and the Web (Supervisor: Prof. Pushpak Bhattacharyya)

- Extended the **M2H2** dataset into **SuM2H2** by adding human-refined multimodal summaries (text, audio, video) explaining why Indian TV scenes are humorous or non-humorous.
- Designed **Muldimorizer**, a multimodal architecture with a novel **Triangulated Attention Layer (TriAL)** to fuse text (BERT), audio (MFCC), and video (CLIP) representations efficiently.
- Achieved **state-of-the-art performance** with an **18% F1 improvement** in humor classification and **103% ROUGE gain** in summarization over strong vision–language baselines.

Industry Experience

Software Engineer II — *Cisco*

Jan 2022–July 2024

- Progressed from *Technical Intern* → *Software Engineer I* → *Software Engineer II* in the Multi-Cloud Management team.
- Designed ETL pipelines between MongoDB and MySQL and integrated AWS, Azure, and GCP account management.
- Developed backend services in Go for a multi-cloud management platform.

Summer Intern — *Fidelity Investments*

May 2021–July 2021

- Gained hands-on experience with SQL query processing and database interactions in a Java-based backend.
- Contributed to integrating and improving SQL query handling within the Java backend system.

Technical Skills

Programming Languages: C, Python, Golang, SQL

Frameworks / Tools: Flask, vLLM, Transformers, Pytorch

Awards and Achievements

- Selected for the **Heidelberg Laureate Forum (HLF)**, 2026.

- Secured **All India Rank 325** among **123,967 candidates** in the Graduate Aptitude Test in Engineering (GATE) — Computer Science, 2025.
- Awarded **Best Bachelor's Thesis (Research Category)** for outstanding undergraduate research.
- Winner, **Young India Innovation Hackathon**, Mysuru Chapter.
- Secured **State Rank 1161** among approximately **200,000 candidates** in the Karnataka Common Entrance Test (KCET), 2018.

Extracurricular Activities

- Volunteer, **AACL-IJCNLP 2025**, Indian Institute of Technology Bombay.
- Member, **Placement Committee**, JSS Science and Technology University, Mysuru (2022).